Supplementary Material

Yield-constrained agrivoltaic potential of India under current and mid-century climate. Sections S1 to S7 give the detail behind Section 2 of the main text, Section S8 shows the validation of the crop and light models, and Section S9 describes the Rajasthan pilot and the Jodhpur soil test that the Discussion cites.

S1. Layout geometry and light model

Table S1. Geometry of the agrivoltaic layouts.

Layout	Row axis	Module tilt / azimuth	Module width per row	Lower edge height	Uncultivated strip	Ground strips simulated
South-facing, elevated	East-west	25° / 180°	2.0 m	3.0 m	None	5
East-west ridge, elevated	North-south	10° / 90° and 270°	2 × 2.0 m	3.0 m	None	3, mirrored to 5
Vertical bifacial	North-south	90° / 90° (bifacial)	2.0 m	0.8 m	0.5 m	Pitch mean

Pitch = module width per row / GCR. Simulated GCR: 0.1–0.5 in steps of 0.1 for all crops, plus 0.6 and 0.7 for wheat and pearl millet and 0.05 for chickpea and groundnut.

For a ground point, beam transmission is the fraction of rays at the projected solar zenith angle that pass between the modules of 81 rows centred on the point, and the sky view factor is the visible fraction of an isotropic sky. Both were evaluated at 200 points across the cultivated part of the pitch and averaged over five strips. The east-west ridge is symmetric about the row axis, so three strips were simulated and mirrored. For vertical rows the strip factors were averaged over the pitch, because the morning and afternoon shade sweeps the whole inter-row space. Hourly irradiance used local solar days in Indian Standard Time and the pvlib Linke turbidity climatology.

Table S2. Light model against the analytical two-dimensional functions in pvlib for single-segment rows (26.4°N, 73.1°E; hourly sun positions of 2010 with solar zenith below 85°).

Layout	GCR	Mean sky view factor, this model	Mean sky view factor, pvlib	Largest pointwise difference	Hourly sunlit fraction, MAE
South-facing	0.1	0.9008	0.9009	0.009	0.0013
South-facing	0.2	0.8040	0.8037	0.009	0.0014
South-facing	0.3	0.7089	0.7086	0.012	0.0015
South-facing	0.4	0.6163	0.6162	0.017	0.0014
South-facing	0.5	0.5274	0.5273	0.016	0.0011
Vertical	0.1	0.9047	0.9050	0.005	0.0014
Vertical	0.2	0.8202	0.8198	0.007	0.0012
Vertical	0.3	0.7435	0.7440	0.009	0.0011
Vertical	0.4	0.6768	0.6770	0.008	0.0011
Vertical	0.5	0.6182	0.6180	0.011	0.0009

The east-west ridge has two segments per row and no direct pvlib counterpart; it uses the same ray-tracing code.

S2. Climate data processing

NEX-GDDP-CMIP6 [1] variables were converted as follows: shortwave radiation from W m^{-2} to MJ m^{-2} d^{-1}; precipitation from kg m^{-2} s^{-1} to mm d^{-1}; dew point from mean daily temperature and relative humidity with the Magnus formula, capped at the daily minimum temperature; and wind speed from 10 m to 2 m with the FAO-56 logarithmic profile [2]. Model calendars of 360 or 365 days were mapped onto the real calendar by nearest-day resampling. The climate grid coincides with the simulation grid, so no spatial interpolation was needed.

Table S3. Climate periods and CO_2 concentration.

Period	Years	Scenario	CO_2 (ppm)	Climate models
Reference	1995–2014	Historical	378	GFDL-ESM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, MRI-ESM2-0, UKESM1-0-LL
Mid-century	2041–2060	SSP2-4.5	507	Same five
Mid-century	2041–2060	SSP5-8.5	563	Same five

Period-mean CO_2 [3], constant within each simulation.

S3. Crop model set-up

DSSAT 4.8 [4,5] was run through the DSSATTools Python interface. Initial soil water was 50% of plant-available water and initial mineral nitrogen was low (0.5 mg NH_4 and 1.0 mg NO_3 per kg soil), and crops were harvested at maturity. Nitrogen was applied as urea incorporated at 5 cm; irrigated wheat received two equal doses at sowing and 30 days after sowing. Rainfed pearl millet and rainfed kharif groundnut were sown automatically in a ±15-day window around the calendar date when soil water in the top 30 cm was 40–100% of plant-available water and soil temperature was 10–40 °C. Soil profiles used the six SoilGrids 2.0 depths [6], sampled at the cell centre; lower limit, drained upper limit, saturation and saturated conductivity came from the Saxton and Rawls functions [7]. The 108 cells (3.6%) without SoilGrids data at the centre took the profile of the nearest valid cell.

Table S4. Management scenarios.

Scenario	Wheat N (kg ha^{-1})	Pearl millet N	Chickpea, groundnut N	Irrigation trigger (share of plant-available water)	Use
Calibrated (main)	30–180 by state	10–30 by state	0	50% (wheat, millet); 20% (legumes)	All periods
Recommended	120 irrigated (60 + 60), 60 rainfed	40	20	50%	Reference period, sensitivity

Triggers apply to irrigated systems, in the top 30 cm of soil.

Table S5. Planting settings.

Crop	Plants m^{-2}	Row spacing (cm)	Sowing depth (cm)	Sowing rule
Wheat	225	22.5	5	GGCMI spring wheat date
Chickpea	33	30	5	Wheat date − 14 d

Crop	Plants m^{-2}	Row spacing (cm)	Sowing depth (cm)	Sowing rule
Pearl millet	11	45	5	GGCMI millet date; rainfed automatic
Groundnut	33	30	5	GGCMI groundnut date; rainfed kharif automatic

Sowing dates from the GGCMI Phase 3 crop calendar [8], irrigated and rainfed calendars separately.

S4. Nitrogen calibration

Simulated open-field cell yields (mean of GFDL-ESM4 and MPI-ESM1-2-HR) were combined into state means weighted by irrigated and rainfed share and by harvested area, and compared with the 1995–2014 mean state yield from the ICRISAT District Level Database [9]. States without enough observed area used the area-weighted median of the calibrated levels. The hill states remained over-estimated at the lowest level; most of their cropland is removed by the slope rule.

Table S6. Calibrated nitrogen level and open-field state yield, reference period.

State	Crop	N (kg ha^{-1})	Observed (kg ha^{-1})	Simulated (kg ha^{-1})
Punjab	Wheat	180	4411	4656
Gujarat	Wheat	180	2553	2854
Haryana	Wheat	90	4172	3739
Rajasthan	Wheat	90	2898	3362
Karnataka	Wheat	90	852	876
Uttar Pradesh	Wheat	60	2761	2652
Madhya Pradesh	Wheat	60	1935	1623
Bihar	Wheat	30	2126	2026
Maharashtra	Wheat	30	1415	1351
West Bengal	Wheat	30	2430	2989
Himachal Pradesh	Wheat	30	1413	3952
Uttarakhand	Wheat	30	1959	3650
Chhattisgarh	Wheat	30	1067	1433
Assam	Wheat	30	1182	1774
Jharkhand	Wheat	30	1421	2281
Other states	Wheat	60	–	–
Gujarat	Pearl millet	30	1355	1129
Haryana	Pearl millet	30	1443	1328
Madhya Pradesh	Pearl millet	30	1406	1173
Uttar Pradesh	Pearl millet	30	1513	1336
Rajasthan	Pearl millet	10	689	637
Maharashtra	Pearl millet	10	758	1183
Karnataka	Pearl millet	10	723	2023
Tamil Nadu	Pearl millet	10	1492	2589
Andhra Pradesh	Pearl millet	10	1277	2059
Other states	Pearl millet	10	–	–

Level 10 kg N ha^{-1} was tested for pearl millet only.

S5. Legume yield-gap factors

Table S7. Observed and simulated open-field legume yields and yield-gap factors (applied to absolute production only).

State	Crop	Observed (kg ha^{-1})	Simulated (kg ha^{-1})	Factor (obs/sim)
Madhya Pradesh	Chickpea	925	854	1.08
Rajasthan	Chickpea	725	522	1.39
Maharashtra	Chickpea	677	436	1.55
Uttar Pradesh	Chickpea	858	1235	0.70
Karnataka	Chickpea	550	1053	0.52
Andhra Pradesh	Chickpea	1096	1292	0.85
Chhattisgarh	Chickpea	746	1089	0.68
Haryana	Chickpea	810	1169	0.69
Gujarat	Chickpea	855	1191	0.72
Telangana	Chickpea	1052	733	1.44
Bihar	Chickpea	1014	820	1.24
Gujarat	Groundnut	1313	2640	0.50
Andhra Pradesh	Groundnut	754	2352	0.32
Karnataka	Groundnut	721	2988	0.24
Tamil Nadu	Groundnut	1979	2654	0.74
Maharashtra	Groundnut	1122	3806	0.30
Rajasthan	Groundnut	1387	2084	0.67
Odisha	Groundnut	1488	3444	0.43
Telangana	Groundnut	1284	1246	1.03
Madhya Pradesh	Groundnut	1176	3665	0.32
Uttar Pradesh	Groundnut	813	3676	0.22

States with at least 50 kha, ordered by area. Area-weighted mean factor: chickpea 1.09, groundnut 0.44.

S6. Aggregation, eligible land and energy

Irrigated and rainfed systems of a crop were combined by production, in the same way as gridded crop-model outputs are aggregated across management systems [8]:

(S1)

where a_w is the share of the crop's area under water regime w and $\bar{Y}_{0,w}$ and RY_w are the open-field yield and RY of that regime. Using the ratio of seasonal means rather than the mean of seasonal ratios in RY keeps seasons with crop failure from dominating. The eligible area of a crop is

(S2)

where A_c is its harvested area (ha) and φ_S the share of the cell with slope ≤ S % (S = 5 in the main case), computed from GLO-90 [10] by finite differences. The rule assumes that cropland is spread evenly over the slope classes of a cell. For single-crop results, capacity is $\eta\, I_{STC}\, A_e\, GCR_{max}$. Annual energy is

(S3)

where Y_{sp} is the specific yield, the final yield of IEC 61724-1 [11] (kWh kW_p^{-1} yr^{-1}), of the cell and layout, interpolated at GCR_{max}. It was computed with pvlib [12] from the same hourly irradiance as the crop simulations, for every fifth year of each GCM (twenty model-years per period). Hourly air temperature followed a cosine between the daily minimum at 06:00 and maximum at 15:00.

Table S8. Parameters of the PV energy model.

Parameter	Value
Module	Bifacial glass-glass, n-type
Efficiency at STC	22%
Bifaciality	0.80
Temperature coefficient of power	−0.30% K^{-1}
Cell temperature	Faiman model [13], pvlib default coefficients, 2 m wind
Irradiance on modules	pvlib infinite sheds, isotropic sky, front and rear
Ground albedo	0.20
DC model	PVWatts [14]
System losses	14%
Inverter efficiency	96%

S7. Implementation and quality control

The workflow was written in Python. Crop simulations, one task per cell, GCM, period and management case, ran on a 12-core cloud virtual machine. Each task wrote its results only on completion, so an interrupted run resumed without duplicates, and DSSAT runs that stopped with an error were retried up to three times. A report every 20 minutes checked three rules and stopped the run on a breach: failed tasks above 2%; median open-field yield of a crop outside a plausible range for India (wheat 800–7000, chickpea 200–3000, groundnut 200–5000 and pearl millet 150–4500 kg ha^{-1}, applied after a full pass over all cells); and more than 5% of irrigated open-field seasons with zero yield.

S8. Validation figures

Figs. S1 to S3 compare the crop model with observed state yields and the light model with an analytical reference and with field measurements. Fig. S1 uses the states with at least 50 kha of the crop. Fig. S2 repeats Table S2 in graphical form. Fig. S3 uses the PPFD measured by Santra et al. [15] at the 105 kW_p agrivoltaic plant of ICAR-CAZRI, Jodhpur, on clear days of 20 to 24 December 2018, 10 cm above the ground. Rows run east-west, modules face south at 26° tilt with the lower edge at 0.5 m, and the three blocks differ in the number of module rows (one, one and a half, two and a half) and in the interspace (2.95, 5.55 and 8.90 m). Measured and modelled values are both relative to the unshaded interspace, which removes the conversion between irradiance and photon flux.

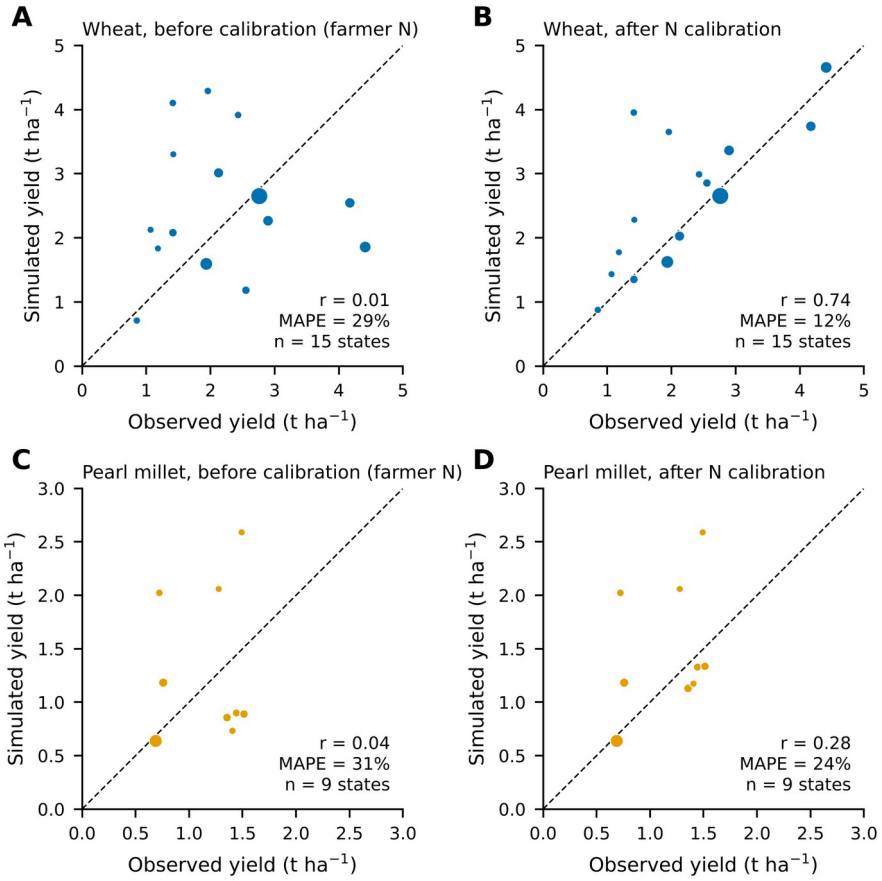


Fig. S1. State mean open-field yields, simulated against observed (ICRISAT District Level Database, 1995–2014), before and after nitrogen calibration. (A, B) Wheat. (C, D) Pearl millet. Before calibration all states used farmer nitrogen (wheat 60, pearl millet 10 kg N ha^{-1}); after calibration each state used its calibrated level (Table S6). Simulated yields are the mean of GFDL-ESM4 and MPI-ESM1-2-HR. Marker area is proportional to observed harvested area; only states with at least 50 kha are shown. The dashed line is 1:1. r is the Pearson correlation and MAPE the mean absolute percentage error weighted by harvested area. Pearl millet in the southern states (Karnataka, Tamil Nadu, Andhra Pradesh) stays over-predicted even at 10 kg N ha^{-1}.

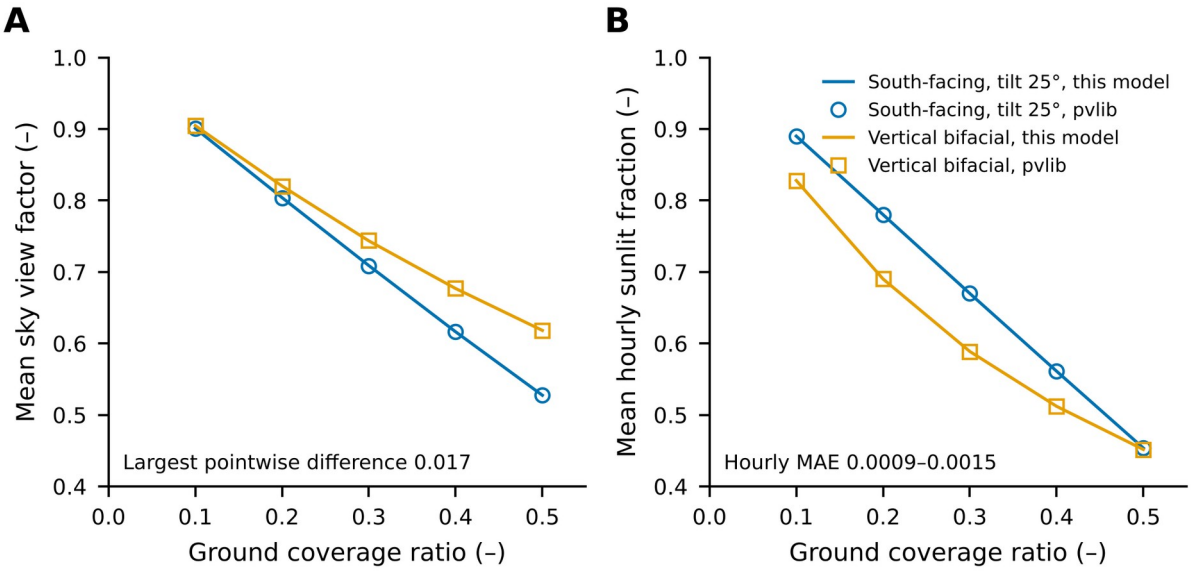


Fig. S2. Light model against the analytical view-factor functions of pvlib for single-segment rows at 26.4°N, 73.1°E (hourly sun positions of 2010, solar zenith below 85°). (A) Sky view factor of the ground averaged across the pitch. (B) Sunlit fraction of the ground averaged over all daylight hours.

Lines are this model and open markers pvlib, for the south-facing layout (tilt 25°) and the vertical bifacial layout. The largest pointwise difference in sky view factor is 0.017 and the mean absolute error of the hourly sunlit fraction is 0.0009 to 0.0015.

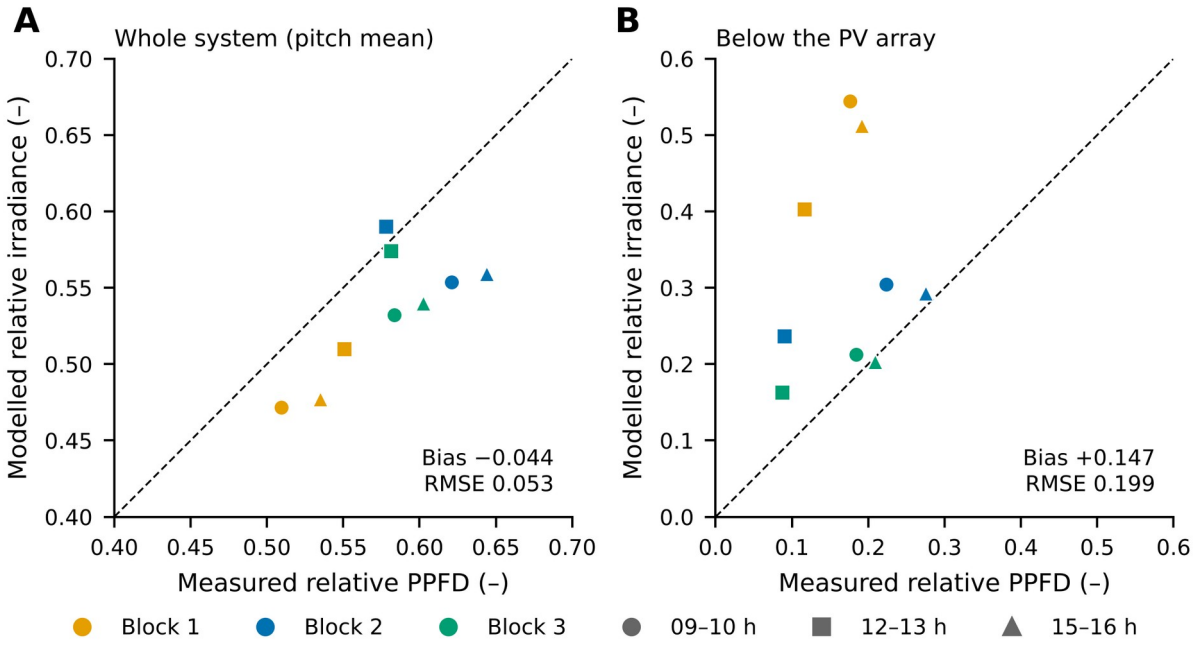


Fig. S3. Modelled relative irradiance against measured relative PPFD at ICAR-CAZRI, Jodhpur [15], by block (colour) and time window (marker). (A) Whole system, averaged over the row pitch: bias −0.044, RMSE 0.053. (B) Ground below the PV array: bias +0.147, RMSE 0.199. The model used clear-sky beam and diffuse irradiance with a broadband diffuse fraction. The dashed line is 1:1. The model reproduces the pitch-mean light but over-estimates the light directly below the modules, most strongly under the single-row Block 1. In late December the noon solar elevation at Jodhpur is about 40°, so below-array light is over-estimated for low winter sun: modelled beam reaches the ground under the low modules more often than measured, and the model includes neither frames, supports nor reflection from the ground and the array. The crop model uses light averaged over strips that span the cultivated pitch, which corresponds to panel A.

S9. Rajasthan pilot and Jodhpur soil test

Before the national run, the light and crop models were tested in a pilot over Rajasthan, the state with the largest pearl millet area and a large wheat and chickpea area. The pilot used 498 cells of the same 0.25° grid, the same five NEX-GDDP-CMIP6 GCMs [1], SoilGrids 2.0 profiles [6], cultivars and light model as the national run, for 1995–2014 (378 ppm CO_2) and 2041–2060 under SSP2-4.5 (507 ppm). It differed from the national run in four ways: nitrogen was not calibrated, and two management cases were run instead (Table S9); sowing dates were fixed (wheat 15 November, chickpea 1 November, groundnut 25 June) except for rainfed pearl millet, which was sown automatically in July; GCR was simulated from 0.1 to 0.5 only, in steps of 0.1; and GCR_{max} was taken as the largest simulated GCR whose RY met the threshold, without interpolation. Chickpea and groundnut were run as rainfed and irrigated systems and, for chickpea, also as their area-weighted mix. The SSP2-4.5 runs used farmer management only. RY was computed per cell and GCM as in the main text (Eq. (2)) and is reported as the median over GCMs and then over cells.

Table S9. Management cases of the Rajasthan pilot.

Case	Wheat	Chickpea, groundnut	Pearl millet
Farmer	60 kg N ha^{-1} (30 at sowing, 30 at 30 d); irrigated at 50% of plant-available water	No N; rainfed, and irrigated at 20% of plant-available water	10 kg N ha^{-1}; rainfed, automatic sowing in July
Recommended	120 kg N ha^{-1} (60 + 60); irrigated at 50%	20 kg N ha^{-1}; irrigated at 50%	40 kg N ha^{-1}; rainfed, automatic sowing in July

Irrigation triggers refer to the top 30 cm of soil.

Table S10 gives the pilot values that the Discussion uses. The soil test was run for the single cell that contains the ICAR-CAZRI experimental site at Jodhpur (26.375°N, 73.125°E), with NASA POWER daily weather for 1995–2014 [16], FAO-56 Penman-Monteith evapotranspiration [2] and the same light model. Rainfed pearl millet received 40 kg N ha^{-1} and was sown automatically in July. The SoilGrids 2.0 profile of the cell was compared with a generic loamy sand profile, to test how strongly soil texture changes open-field yield and RY.

Table S10. Rajasthan pilot and Jodhpur soil test: values cited in the main text.

Quantity	Case	Value
Wheat RY, south-facing, GCR 0.3	Farmer / recommended	1.13 / 0.90
Wheat RY, south-facing, GCR 0.5	Farmer / recommended	0.86 / 0.51
Pearl millet RY, south-facing, GCR 0.5	Farmer / recommended	0.89 / 0.78
GCR_{max} at RY ≥ 0.80, chickpea systems	Farmer, all layouts	0.1–0.2
GCR_{max} at RY ≥ 0.80, groundnut systems	Farmer, all layouts	0.2–0.4
GCR_{max} at RY ≥ 0.80, wheat and pearl millet	Farmer, all layouts	0.5 (largest simulated)
GCR_{max} at RY ≥ 0.80, wheat	Recommended; south / east-west / vertical	0.3 / 0.5 / 0.5
Largest median change in RY, SSP2-4.5 vs 1995–2014	Farmer, any system, layout, GCR	0.04
Jodhpur open-field pearl millet yield (kg ha^{-1})	SoilGrids / loamy sand	886 / 583
Jodhpur pearl millet RY, south-facing, GCR 0.3	SoilGrids / loamy sand	1.02 / 0.87

Pilot: median over five GCMs and then over cells. Jodhpur: mean over the simulated seasons.

S10. Results with calibrated nitrogen alone

The main text reports the robust nitrogen case. Figs. S4 and S5 show the same results with calibrated nitrogen alone, the upper bound of Section 3.4 (east-west layout, $\tau = 0.80$: 45.5 TWp against 39.2 TWp in the robust case).

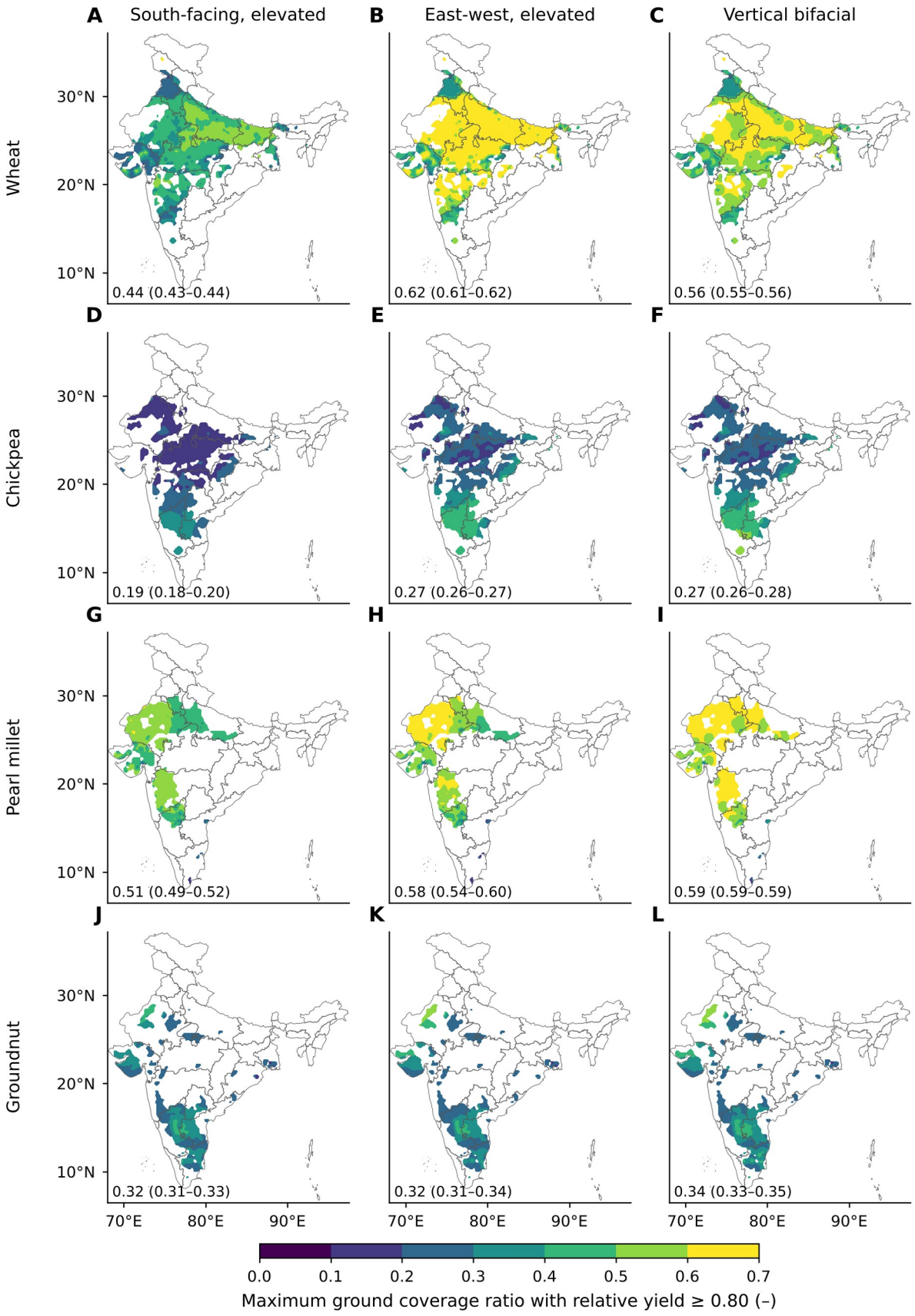


Fig. S4. As Fig. 3, with calibrated nitrogen alone (upper bound): largest ground coverage ratio (GCR_{max}) that keeps relative yield at or above 0.80, by crop (columns) and layout (rows), 1995–2014; vertical GCR_{max} capped at 0.6. Cell values are the median over five GCMs, interpolated for display; the number in each panel is the area-weighted mean with the GCM range. Boundaries: DataMeet India community, States/Admin2 (CC BY 4.0).

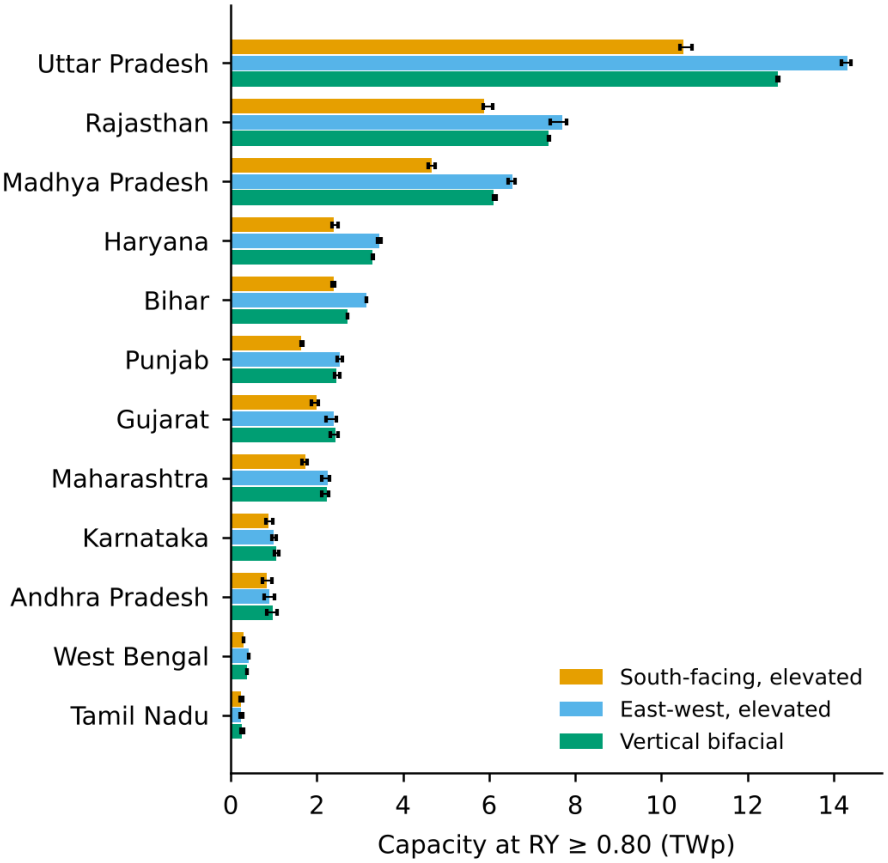


Fig. S5. As Fig. 6, with calibrated nitrogen alone (upper bound): capacity of the leading states at RY ≥ 0.80 by layout, 1995–2014, slope ≤ 5%, vertical GCR_{max} capped at 0.6. Bars are the five-GCM median and error bars the GCM range.

S11. Land-supply curve

Fig. S6 shows how much cropland a given APV capacity needs if the cells with the highest yield-constrained power density are used first (east-west layout, RY ≥ 0.80, slope ≤ 5%, robust nitrogen). For each GCM the cells are sorted by capacity per hectare and capacity and land are accumulated; the curve is the five-GCM median. At this order, 300 GWp needs 0.21 Mha and 500 GWp 0.35 Mha, against 0.30 and 0.51 Mha at the national mean density.

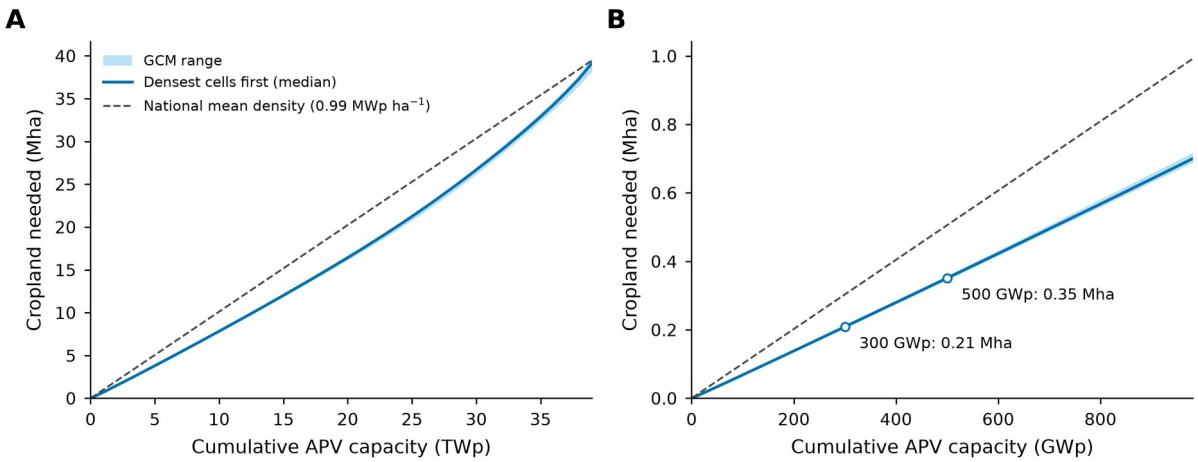


Fig. S6. Cropland needed for a given APV capacity when the densest cells are used first, east-west layout, RY ≥ 0.80, slope ≤ 5%, 1995–2014. (A) Full range; (B) first 1000 GWp. Solid line: five-GCM median; band: GCM range; dashed line: land at the national mean density.

S13. Maps by crop and layout

Fig. S7 shows the largest ground coverage ratio for all four crops and all three layouts in the robust nitrogen case, the maps of which only the east-west layout is shown in the main text (Fig. 3). Figs. S8 and S9 show the change in GCR_{max} by 2041–2060 under SSP2-4.5 and SSP5-8.5, with calibrated nitrogen, for every crop and layout.

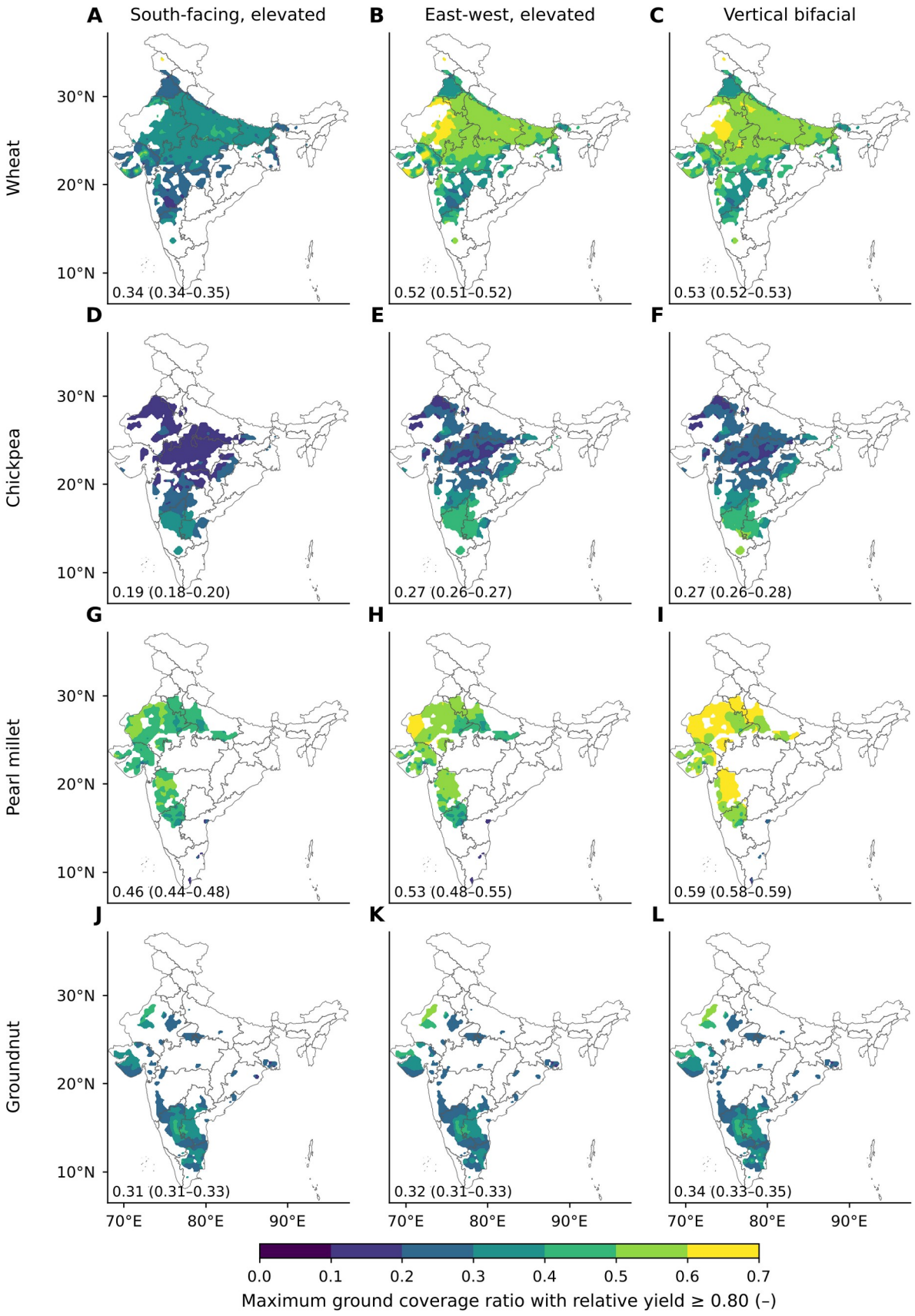


Fig. S7. As Fig. 3, for all crops and layouts: largest ground coverage ratio (GCR_{max}) that keeps relative yield at or above 0.80, by crop (rows) and layout (columns), 1995–2014, robust nitrogen case; vertical GCR_{max} capped at 0.6. Cell values are the median over five GCMs, interpolated for display; the number in each panel is the eligible-area-weighted mean with the GCM range. Boundaries: DataMeet India community, States/Admin2 (CC BY 4.0).

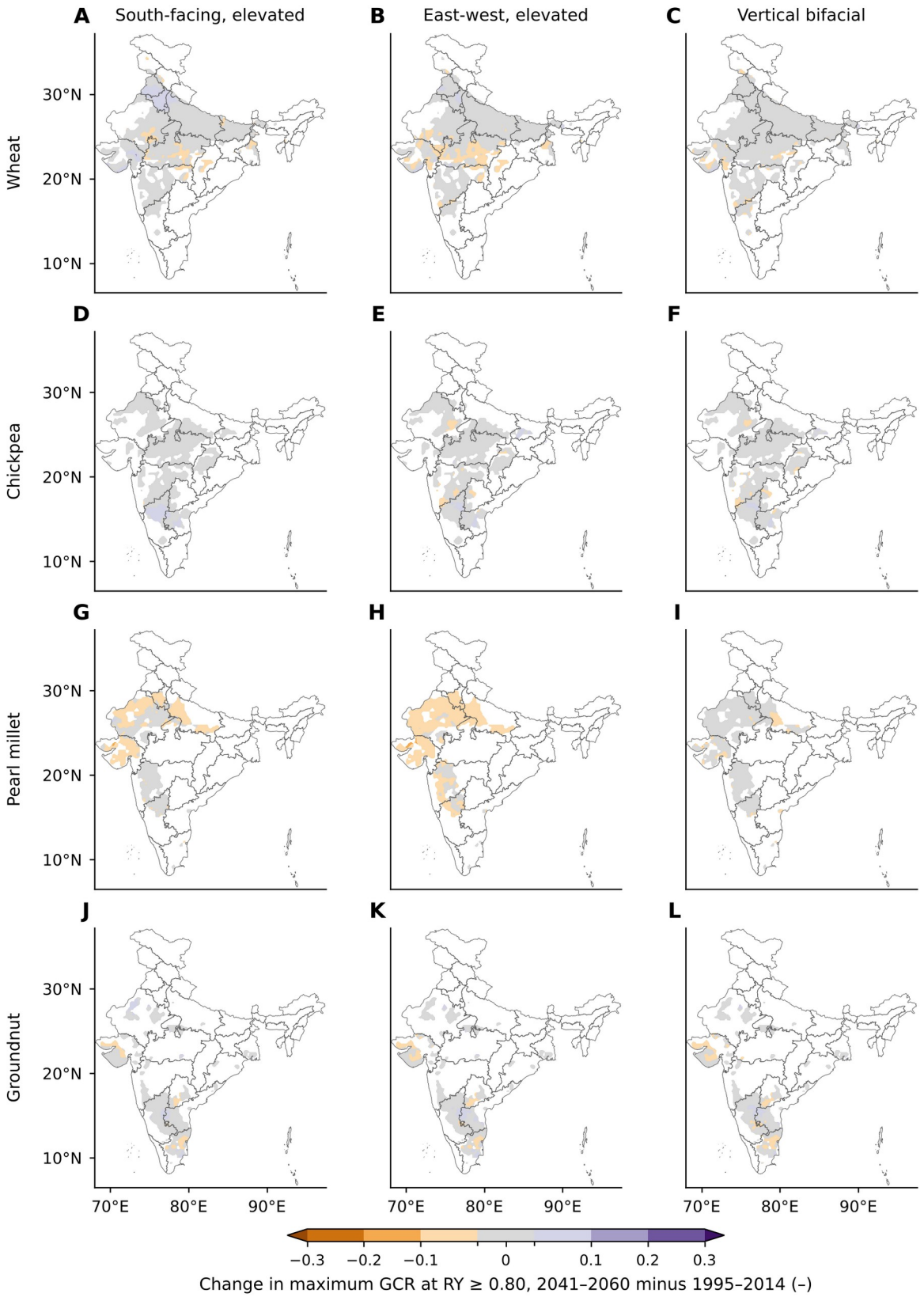


Fig. S8. Change in GCR_{max} at RY ≥ 0.80, 2041–2060 under SSP2-4.5 minus 1995–2014, by crop and layout (five-GCM median, interpolated for display; calibrated nitrogen).

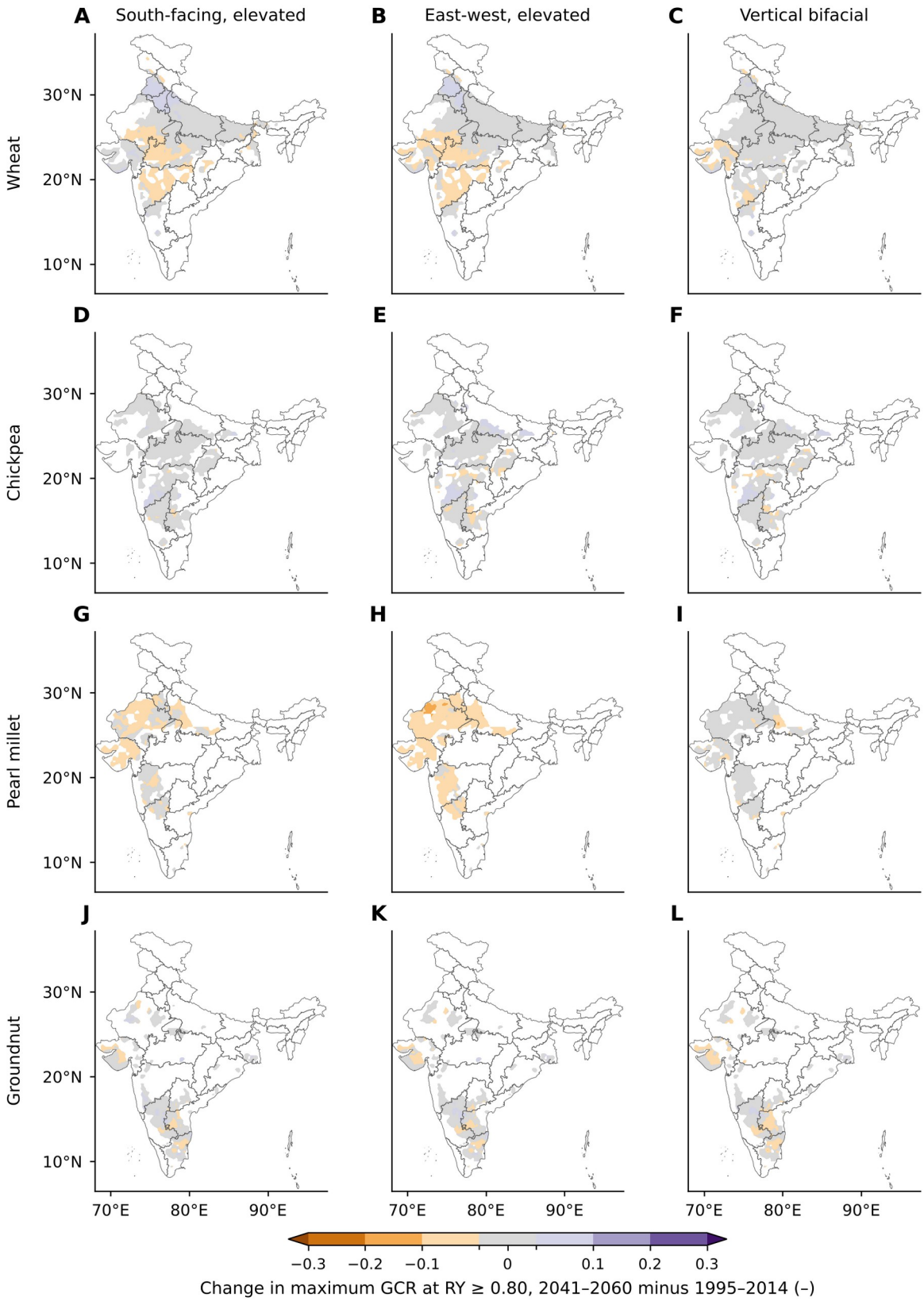


Fig. S9. As Fig. S8, for SSP5-8.5.

S14. Simulated relative yield against reduction in ground light

Fig. S10 plots the simulated relative yield (RY) of each crop in the east-west layout against the mean reduction in ground light, with calibrated and with recommended nitrogen. The reduction

is one minus the season-mean, strip-mean ground-light factor of the light model, computed for a representative central-Indian site (25° N, 78° E) with a rabi season for wheat and chickpea and a kharif season for pearl millet and groundnut. For the east-west layout the mean reduction is close to the GCR (a GCR of 0.2 gives a reduction of 0.20, 0.5 gives 0.49). Curves are medians over all cells, GCMs and water systems of the 1995–2014 runs; the proportional-loss line marks RY = 1 minus the reduction.

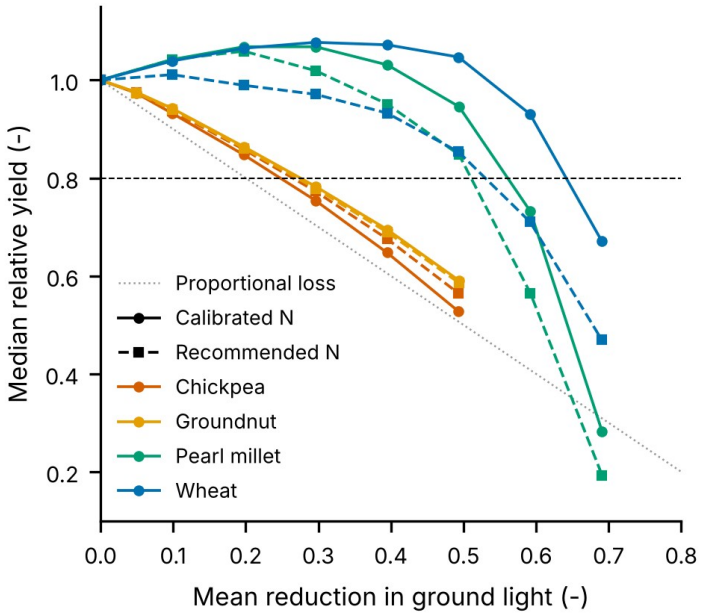


Fig. S10. Simulated median RY against mean reduction in ground light, east-west layout, 1995–2014. Solid lines with circles: calibrated nitrogen; dashed lines with squares: recommended nitrogen; dotted line: proportional loss; dashed horizontal line: RY = 0.80.

Fig. S11 compares these curves with the observations compiled by Jha et al. [17] (public data set of 547 observations from 66 studies; relative yield and reduction in solar radiation, mostly from artificial shading and intercropping trials). Wheat observations are wheat and durum wheat; grain legumes are all grain-legume observations up to a 50% reduction (soybean dominates, with chickpea, mungbean, peanut and bean), compared with the mean of the simulated chickpea and groundnut curves; maize is compared with the simulated pearl millet curve because the compilation contains no millet. Simulated RY was interpolated at each observed reduction. The mean bias (simulated minus observed) with recommended nitrogen was +0.09 for wheat, +0.09 for grain legumes and +0.20 for maize, and +0.16, +0.08 and +0.26 with calibrated nitrogen. The simulated crops are therefore more tolerant of shade than the compiled field data, and the comparison is not an India-specific validation.

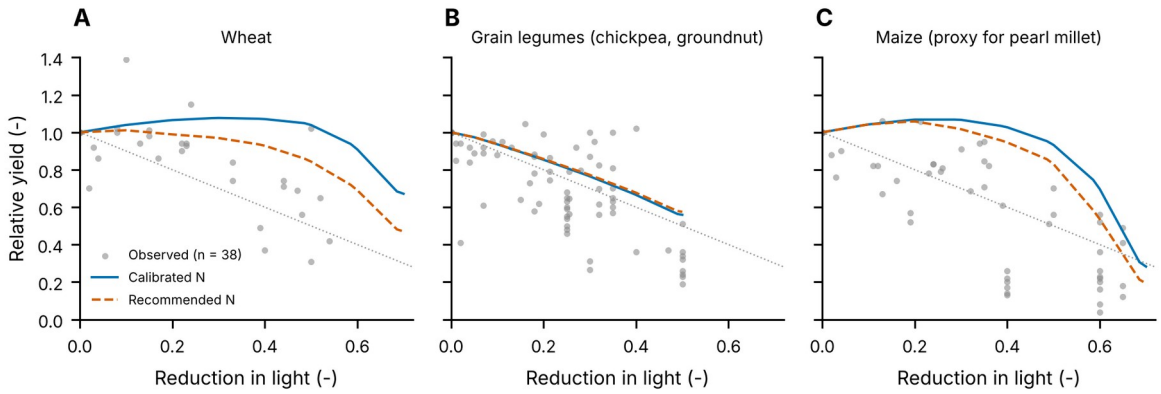


Fig. S11. Observed relative yield against reduction in solar radiation (grey points, Jha et al.) and simulated median RY against reduction in ground light (lines), east-west layout. A, wheat; B, grain legumes (simulated: mean of chickpea and groundnut); C, maize (simulated: pearl millet). Blue solid line: calibrated nitrogen; orange dashed line: recommended nitrogen; dotted line: proportional loss.

Field-response scenario. For each crop we fitted a decreasing (isotonic) curve to the observed RY against the reduction in solar radiation, anchored at RY = 1 for no reduction and with gains above one not credited (wheat from wheat and durum wheat, chickpea and groundnut from all grain legumes, pearl millet from maize). The simulated RY of every cell, GCM, layout and GCR was multiplied by c = min(1, observed RY / national median simulated RY), evaluated at the mean ground-light reduction of that layout and GCR (light model at 25° N, 78° E and the season of the crop). Spatial differences between cells are kept and only the average response is aligned with the observations. The potential was then recomputed as in the main case, with the lower GCR_{max} of the calibrated and recommended nitrogen runs. With τ = 0.80 and slopes up to 5%, the east-west potential is 22.6 TWp (42.1 PWh yr^{-1}), the south-facing 17.1 TWp (34.4 PWh yr^{-1}) and the vertical 23.9 TWp (33.3 PWh yr^{-1}); at τ = 0.66 and 0.90 the east-west potential is 31.6 and 12.8 TWp. The scenario treats light as the only driver, uses one representative site for the light reduction and relies on maize as the proxy for pearl millet, so it is a bound and not a best estimate. The choice of observation set matters little: leaving soybean out of the grain-legume set gives 22.9 TWp, using all C3 cereals for wheat gives 22.3 TWp, and using maize together with C3 cereals for pearl millet gives 23.6 TWp, against 22.6 TWp for the main set (east-west, 80% threshold, slope ≤5%).

Supplementary references

[1] Thrasher B, Wang W, Michaelis A, Melton F, Lee T, Nemani R. NASA Global Daily Downscaled Projections, CMIP6. Sci Data 2022;9:262. https://doi.org/10.1038/s41597-022-01393-4.

[2] Allen RG, Pereira LS, Raes D, Smith M. Crop evapotranspiration: Guidelines for computing crop water requirements. FAO Irrigation and Drainage Paper 56. Rome: FAO; 1998. https://www.fao.org/4/x0490e/x0490e00.htm.

[3] Meinshausen M, Nicholls ZRJ, Lewis J, Gidden MJ, Vogel E, Freund M, et al. The shared socio-economic pathway (SSP) greenhouse gas concentrations and their extensions to 2500. Geosci Model Dev 2020;13:3571–605. https://doi.org/10.5194/gmd-13-3571-2020.

[4] Jones JW, Hoogenboom G, Porter CH, Boote KJ, Batchelor WD, Hunt LA, et al. The DSSAT cropping system model. Eur J Agron 2003;18:235–65. https://doi.org/10.1016/S1161-0301(02)00107-7.

[5] Hoogenboom G, Porter CH, Boote KJ, Shelia V, Wilkens PW, Singh U, et al. Decision Support System for Agrotechnology Transfer (DSSAT) Version 4.8. Gainesville, FL: DSSAT Foundation; 2021. https://dssat.net.

[6] Poggio L, de Sousa LM, Batjes NH, Heuvelink GBM, Kempen B, Ribeiro E, et al. SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. SOIL 2021;7:217–40. https://doi.org/10.5194/soil-7-217-2021.

[7] Saxton KE, Rawls WJ. Soil water characteristic estimates by texture and organic matter for hydrologic solutions. Soil Sci Soc Am J 2006;70:1569–78. https://doi.org/10.2136/sssaj2005.0117.

[8] Jägermeyr J, Müller C, Ruane AC, Elliott J, Balkovic J, Castillo O, et al. Climate impacts on global agriculture emerge earlier in new generation of climate and crop models. Nat Food 2021;2:873–85. https://doi.org/10.1038/s43016-021-00400-y.

[9] Swamikannu N, Pingali PR, Padmanabhan J, Ravula P, Gumma MK, Mohammed IA, et al. District Level Data (DLD) and Dashboard for Agriculture and Allied-sectors in India, version 1. ICRISAT Dataverse; 2021. https://doi.org/10.21421/D2/XFB1BZ.

[10] European Space Agency, Airbus. Copernicus DEM, global 90 m (GLO-90). 2022. https://doi.org/10.5270/ESA-c5d3d65.

[11] IEC 61724-1:2021. Photovoltaic system performance – Part 1: Monitoring. Geneva: International Electrotechnical Commission; 2021. https://webstore.iec.ch/en/publication/65561.

[12] Anderson KS, Hansen CW, Holmgren WF, Jensen AR, Mikofski MA, Driesse A. pvlib python: 2023 project update. J Open Source Softw 2023;8:5994. https://doi.org/10.21105/joss.05994.

[13] Faiman D. Assessing the outdoor operating temperature of photovoltaic modules. Prog Photovolt Res Appl 2008;16:307–15. https://doi.org/10.1002/pip.813.

[14] Dobos AP. PVWatts Version 5 Manual. NREL/TP-6A20-62641. Golden, CO: National Renewable Energy Laboratory; 2014. https://doi.org/10.2172/1158421.

[15] Santra P, Meena HM, Yadav OP. Spatial and temporal variation of photosynthetic photon flux density within agrivoltaic system in hot arid region of India. Biosyst Eng 2021;209:74–93. https://doi.org/10.1016/j.biosystemseng.2021.06.017.

[16] NASA Langley Research Center. Prediction Of Worldwide Energy Resources (POWER) Project, daily and climatology data. Hampton, VA: NASA. https://power.larc.nasa.gov [accessed November 2026].

[17] Jha A, Heiser G, Kelvey R, Huang Q. Crop yield responses to reduced solar radiation in agrivoltaic systems: crop-specific patterns and shading thresholds. Agronomy 2026;16:985. https://doi.org/10.3390/agronomy16100985.